

outcomes $X_1, X_2, \dots, X_n$ (these outcomes form a set of mutually exclusive and exhaustive events). The expected value of X conditional on S is the first outcome, X_1 , times the probability of the first outcome given S , $P(X_1 | S)$, plus the second outcome, X_2 , times the probability of the second outcome given S , $P(X_2 | S)$, and so forth, as follows:

$$E(X | S) = P(X_1 | S)X_1 + P(X_2 | S)X_2 + \dots + P(X_n | S)X_n. \quad (4)$$

We will illustrate this equation shortly.

Parallel to the total probability rule for stating unconditional probabilities in terms of conditional probabilities, there is a principle for stating (unconditional) expected values in terms of conditional expected values. This principle is the **total probability rule for expected value**.

Total Probability Rule for Expected Value

The formula follows:

$$E(X) = E(X | S)P(S) + E(X | S^C)P(S^C), \quad (5)$$

where S^C is the “complement of S ,” which means event or scenario “ S ” does not occur.

$$E(X) = E(X | S_1)P(S_1) + E(X | S_2)P(S_2) + \dots + E(X | S_n)P(S_n), \quad (6)$$

where $S_1, S_2, \dots, S_n$ are mutually exclusive and exhaustive scenarios or events.

The general case, Equation 6, states that the expected value of X equals the expected value of X given Scenario 1, $E(X | S_1)$, times the probability of Scenario 1, $P(S_1)$, plus the expected value of X given Scenario 2, $E(X | S_2)$, times the probability of Scenario 2, $P(S_2)$, and so forth.

To use this principle, we formulate mutually exclusive and exhaustive scenarios that are useful for understanding the outcomes of the random variable. This approach was employed in developing the probability distribution of BankCorp’s EPS in Example 1, as we now discuss.

EXAMPLE 2

BankCorp’s Earnings per Share, Part 2

The earnings of BankCorp are interest rate sensitive, benefiting from a declining interest rate environment. Suppose there is a 0.60 probability that BankCorp will operate in a *declining interest rate environment* in the current fiscal year and a 0.40 probability that it will operate in a *stable interest rate environment* (assessing the chance of an increasing interest rate environment as negligible). If a *declining interest rate environment* occurs, the probability that EPS will be USD2.60 is estimated at 0.25, and the probability that EPS will be USD2.45 is estimated at 0.75. Note that 0.60, the probability of *declining interest rate environment*, times 0.25, the probability of USD2.60 EPS given a *declining interest rate environment*, equals 0.15, the (unconditional) probability of USD2.60 given in the table in Exhibit 1. The probabilities are consistent. Also, $0.60(0.75) = 0.45$, the probability of USD2.45 EPS given in Exhibit 1. The **probability tree diagram** in Exhibit 2 shows the rest of the analysis.